

LABLAB X AMD AI ACADEMY CHALLENGE

Circuit OS

An AI hardware compiler. Plain English in — validated schematic, firmware, simulation and BOM out.

AMD ROCm

AMD Developer Cloud

vLLM

Open source • AGPL-3.0

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github.com/monickverma/cursor-electronics

THE PROBLEM

LLMs are confidently wrong about hardware

- ▶ They **invent pin numbers** and library functions that don't exist
- ▶ They leave nodes floating and **forget I²C pull-ups** or RS-485 termination
- ▶ A wrong value — **1nF instead of 100nF** — looks fine until the board is built
- ▶ Beginners can't spot the mistake; engineers waste hours reviewing

THE ONE RULE

The AI never writes SPICE, KiCad or firmware



The model only fills a schema. Deterministic compilers do the rest — **same input, same output.**

WHAT YOU GET

From one sentence to a buildable design

Validated IR

Floating nodes, voltage ratings, pull-ups, RS-485 termination and PWM pins checked by a rule engine.

Simulation

ngspice via Celery + Redis, graded pass/fail at 15% tolerance. Wrong parts caught before building.

Firmware

Jinja2-templated Arduino code that compiles for a real Uno.

Patches

“Make the LED brighter” changes only what was asked — the rest stays intact.

Explanations

Consequences, not descriptions: what breaks if you change R3.

BOM + KiCad

Bill of materials and a schematic, plus an experimental A* PCB router.

BUILT ON AMD

Runs on open models, served by vLLM on ROCm

- ▶ `AI_PROVIDER=openai_compat` switches the whole AI layer to **vLLM on AMD GPUs**
- ▶ Forced tool calls become **guided decoding** — output always matches the schema
- ▶ The validate-and-retry loop is what makes a **7B open model usable** for hardware design
- ▶ `scripts/amd_benchmark.py` measures valid-IR rate, retries and latency on the GPU

```
$ vllm serve Qwen/Qwen2.5-7B-Instruct --enable-auto-tool-choice --tool-call-parser hermes
$ AI_PROVIDER=openai_compat python scripts/amd_benchmark.py
```

STATUS · V0.1.0

Engineered, tested, open

430+

passing tests

5

templates: DHT22 alert,
MAX485 Modbus RTU, LED,
RC filter, voltage divider

2

AI backends: Claude, or open
models on AMD ROCm

JWT auth · rate limiting · PostgreSQL · FastAPI · Next.js · AGPL-3.0

NEXT

Where Circuit OS goes from here

- ▶ **Free-form circuits** beyond the five templates
- ▶ **Fine-tune an open model on AMD GPUs** using validated IRs and retry corrections as training data
- ▶ Publish Claude vs. open-model-on-ROCM benchmarks
- ▶ Promote the PCB router from experimental to tested